

Long-Only Equity Capacity Run Report

Illustrative Portfolio Analysis

Distributional Migration and Alpha-Decay Model

Run timestamp: 25 September 2026, 21:47:31 BST

Abstract

This is a portfolio-specific run report, separate from the stable methodology. It records the inputs, policy assumptions, fitted parameters, diagnostics, scenario results and capacity thresholds for the illustrative portfolio. It should be regenerated for each portfolio or input snapshot.

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1 Executive summary

The current portfolio has £5bn AUM, 4% expected gross relative return, 4% tracking error and 80% one-way turnover. The fitted model reconciles current expected impact to 8.91 bp per executed notional, implying 14.26 bp annual impact drag and net IR of 0.964.

At £20bn, the model estimates that average execution time rises from 1.81 to 3.21 days, 95.67% of current gross alpha is captured after delay, and net IR falls to 0.870. The illustrative net-IR threshold of 0.75 is crossed at approximately £45.62bn.

Run classification: Illustrative aggregate-bucket analysis. Results are scenarios, not confidence bounds. Trade-level alpha weights, observed execution horizons and empirically estimated signal half-lives were not provided.

2 Run identity and provenance

Field	Value
Run ID	capacity-20260925-214731
Model	Distributional long-only capacity model
Python environment	Conda environment pyx86
Model source	long_only_capacity_tool.py
Input type	Illustrative ADV buckets, executed-notional weighted
Distribution	Regularised Burr XII
Output timestamp	2026-09-25 21:47:31 BST

3 User-supplied strategy inputs

Table 1: Strategy snapshot

Input	Value
Current AUM	£5bn
Gross expected relative return	4.00% p.a.
Tracking error	4.00% p.a.
Gross information ratio	1.00
One-way turnover	80% p.a.
Average holdings	100
Annual executed notional	£8bn

Table 2: Liquidity and impact inputs

ADV bucket	Representative	Notional share	Expected bp	Realised bp
0–1%	0.5%	20%	2.0	2.2
1–5%	2.5%	35%	4.0	4.4
5–10%	7.5%	20%	7.0	7.8
10–50%	25%	15%	14.0	16.0
50–100%	70%	7%	28.0	34.0
>100%	125%	3%	55.0	72.0

4 Policy and scenario parameters

Table 3: User-set modelling assumptions

Parameter	Run value
Participation-scale elasticity η	0.85
Tail-thickening elasticity κ	0.00
Target daily participation ρ	10% ADV
Maximum execution horizon $D_{\max}$	10 trading days
Alpha half-life H_α	10 trading days
AUM curve	£1bn to £100bn
Net IR threshold	0.75
Alpha-retention threshold	80%
Annual impact-drag threshold	100 bp
Alpha-capture threshold	95%

5 Fitted distribution and impact parameters

The regularised Burr XII fit produced

$$c = 0.9437, \quad d = 1.3245, \quad \lambda_0 = 0.0590, \quad cd = 1.25.$$

The expected impact fit produced approximately

$$a = 3.5252, \quad b = 39.5867, \quad \gamma = 1.1105.$$

The current distributional expected impact was rescaled to the observed weighted bucket cost of 8.91 bp.

Table 4: Distribution fit at current AUM

ADV bucket	Observed share	Fitted share
0–1%	20.00%	20.33%
1–5%	35.00%	35.55%
5–10%	20.00%	16.54%
10–50%	15.00%	21.72%
50–100%	7.00%	3.20%
>100%	3.00%	2.66%

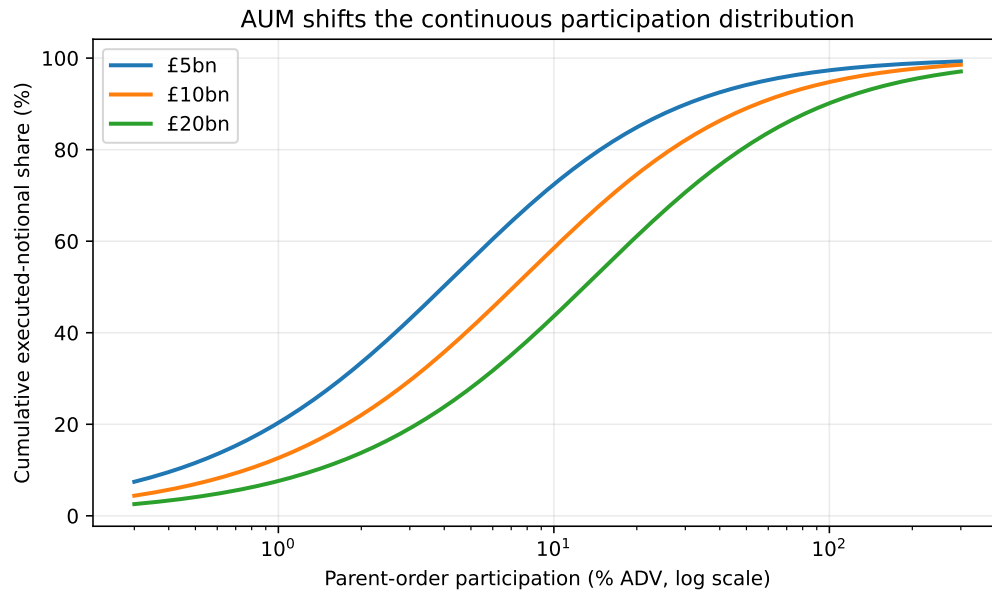


Figure 1: Fitted participation distribution at three AUM levels.

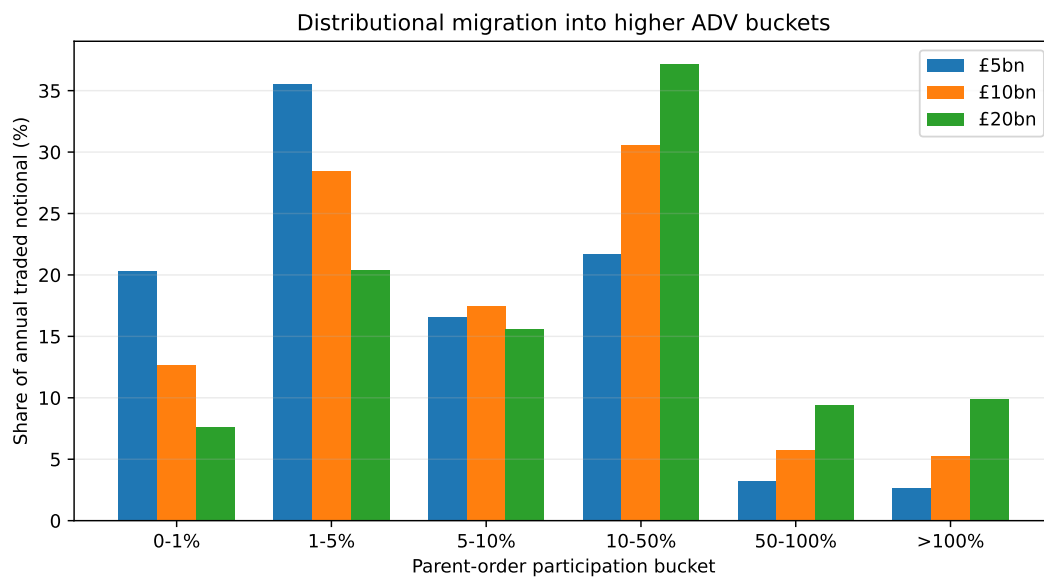


Figure 2: Smooth migration of executed notional across participation buckets.

6 AUM scenario results

Table 5: Capacity scenarios

Metric	£5bn	£10bn	£20bn
Gross alpha before delay	4.000%	4.000%	4.000%
Alpha captured	100.00%	98.17%	95.67%
Gross alpha after delay	4.000%	3.927%	3.827%
Weighted expected impact	8.91 bp	13.22 bp	21.60 bp
Annual impact drag	14.26 bp	21.16 bp	34.56 bp
Net alpha	3.857%	3.715%	3.481%
Net IR	0.964	0.929	0.870
Average execution days	1.81	2.39	3.21
Notional requiring multiple days	27.58%	41.48%	56.40%
Notional requiring at least 3 days	15.13%	25.40%	38.90%

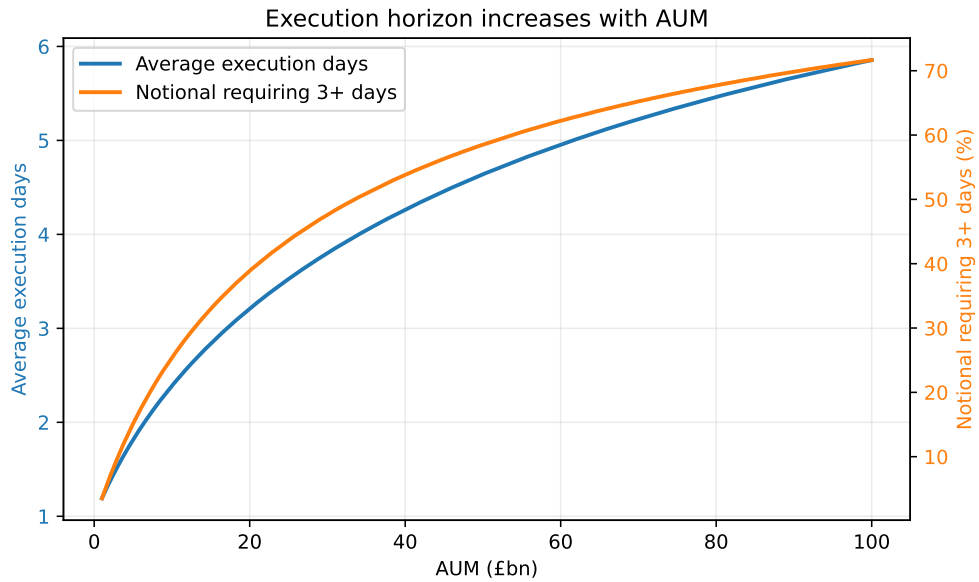


Figure 3: Execution horizon and multi-day notional as AUM increases.

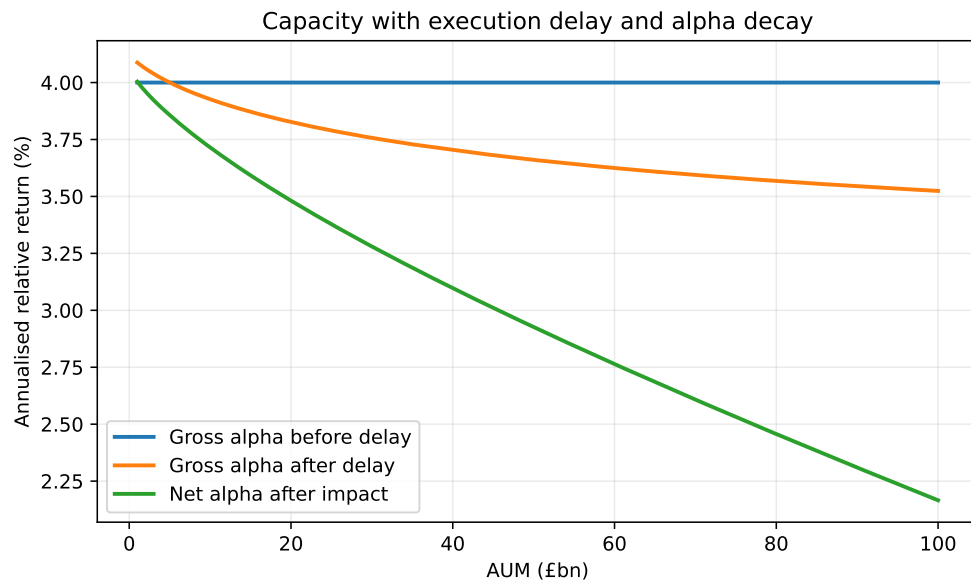


Figure 4: Alpha before delay, after delay and after expected impact.

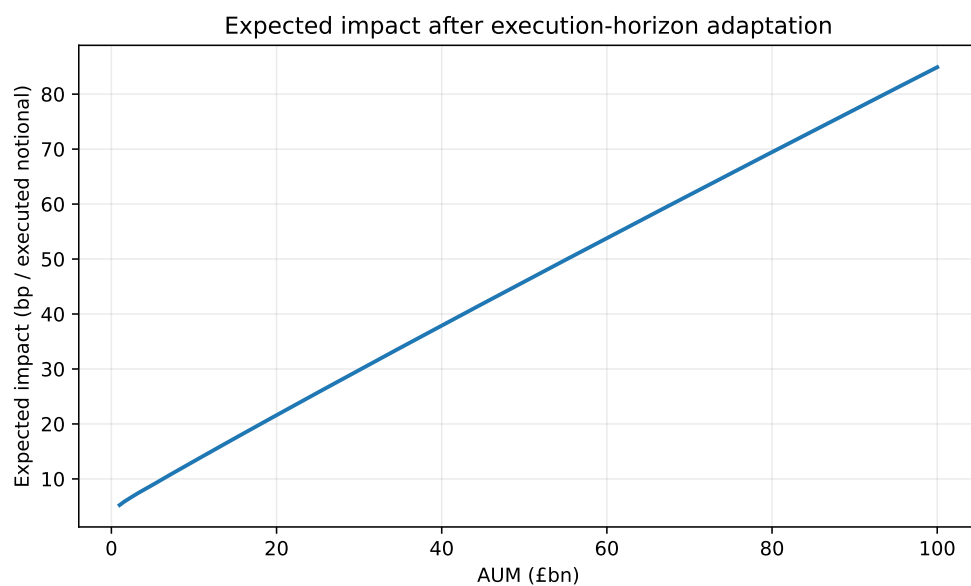


Figure 5: Expected impact after execution-horizon adaptation.

7 Capacity thresholds

Table 6: Exact threshold crossings

Threshold	AUM crossing
Alpha capture $\leq 95\%$	£23.51bn
Alpha retained after delay and impact $\leq 80\%$	£34.29bn
Net IR ≤ 0.75	£45.62bn
Annual impact drag ≥ 100 bp	£71.08bn

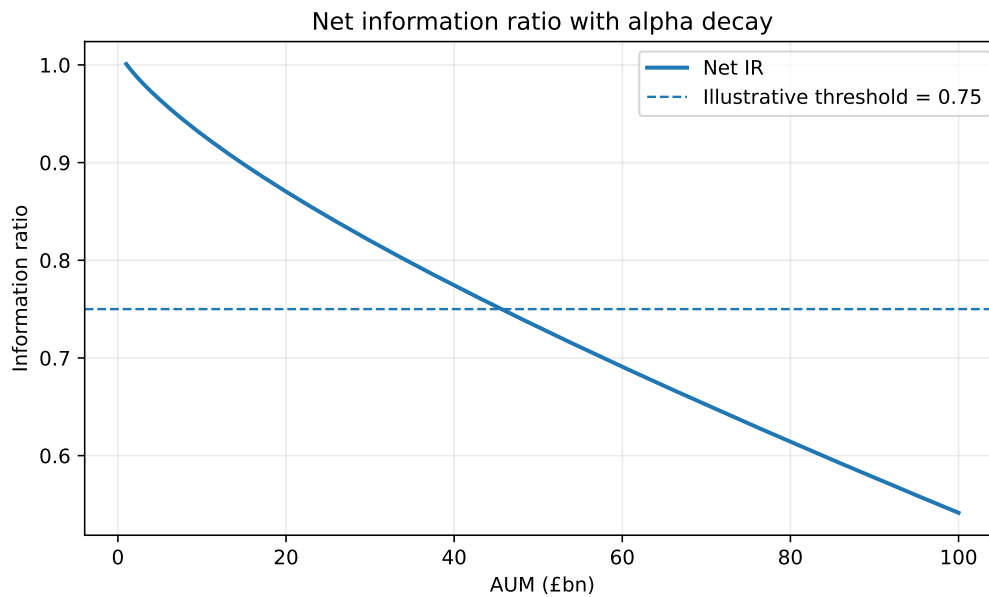


Figure 6: Net information ratio and illustrative 0.75 threshold.

8 Validation results

Check	Status	Detail
Input bucket shares sum to 100%	Pass	Exact within numerical tolerance.
Fitted bucket shares sum to 100%	Pass	CDF differences cover the positive support.
Current expected impact reconciliation	Pass	Model equals supplied 8.91 bp.
Alpha capture at current AUM	Pass	Normalised to 100%.
Positive AUM and half-life	Pass	Both validation constraints satisfied.
Monotone alpha capture versus AUM	Pass	Capture declines over the evaluated grid.
Trade-level alpha weights available	Warning	Alpha is assumed proportional to executed notional.
Empirical execution horizons available	Warning	Horizon is generated by the 10% ADV policy rule.
Tail estimate based on rich sample	Warning	Six aggregate buckets require regularisation.

9 Interpretation and limitations

The run indicates that alpha decay becomes economically visible before annual impact drag reaches 100 bp. Under the selected parameters, the 95% alpha-capture threshold occurs near £23.5bn, while net IR reaches 0.75 near £45.6bn. This ordering is specific to a ten-day alpha half-life and a 10% ADV daily execution target.

The results should not be read as a recommended fund-size limit without sensitivity analysis. The largest uncertainties are the AUM elasticity, signal half-life, daily participation policy, aggregate tail fit and assumption that notional share proxies for alpha share.

10 Run outputs

The Python run also produces CSV files for the strategy summary, distribution fit, AUM scenarios, bucket migration, full capacity curve and exact thresholds, together with the figures embedded in this report. A production run should archive these files with the portfolio snapshot and model version so the report can be reproduced.